

Tiktok Engagement

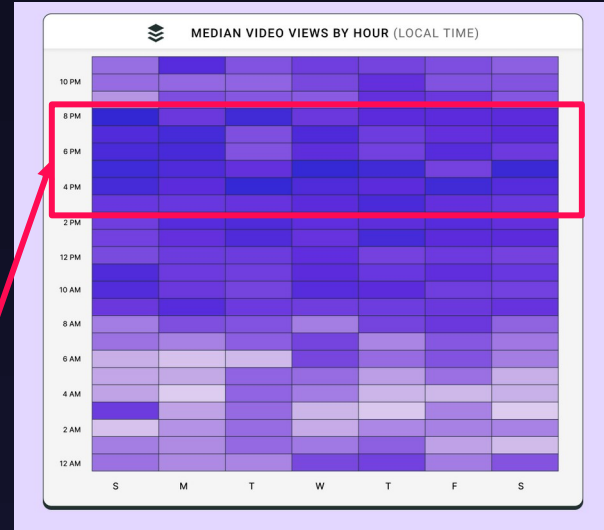
Regression Analysis On
Posting-Time and
Engagement

Presented by Team B3
Trave.LA.rs



Problem Statement

- Digital platforms shape creator and business success.
- Independent research done by *Buffer* (analyzed 1 million video sent to them), **evening time had higher video views by hour**



Research Question: Does posting-time influence TikTok engagement?

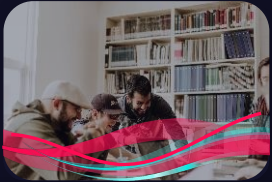
Null Hypothesis: There is no relationship between posting in evening time and engagement

Alternative Hypothesis: Posting in the evening drives higher engagement than other timeframes.

WHY Analyze with Data?



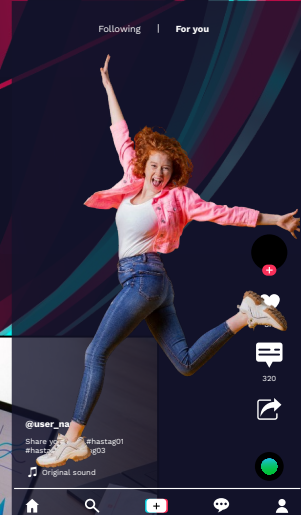
Intuition about “Best Posting Times” is often wrong



Engagement is shaped by many competing factors

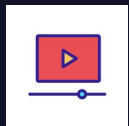


TikTok performance is highly unpredictable

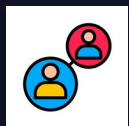


Our Dataset

North America:



2912 Videos



668 Creators

Main Independent Variable:
Publish Period

Main Dependent Variable:
Engagement per 1K

Potential Secondary Variables:

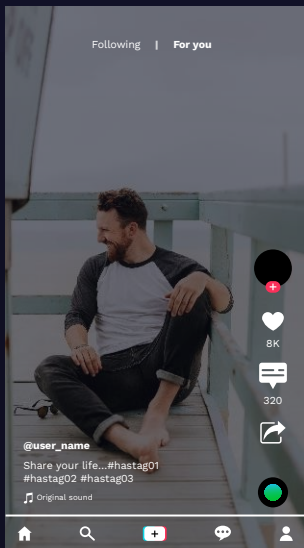
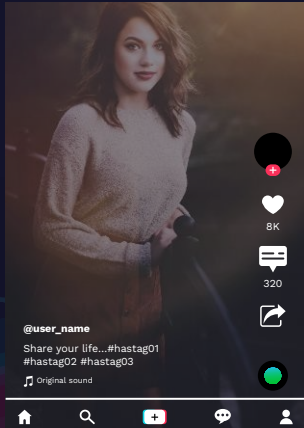
- 1) Sound Type
- 2) Traffic Source
- 3) Creator Average Views

Ideal Experiment

1. Random creators posts many similar-quality videos.
2. Each video is randomly assigned to a publish period (morning, afternoon, evening, late night).
3. All other factors remain constant.
4. Engagement outcomes are then compared across randomized time conditions.

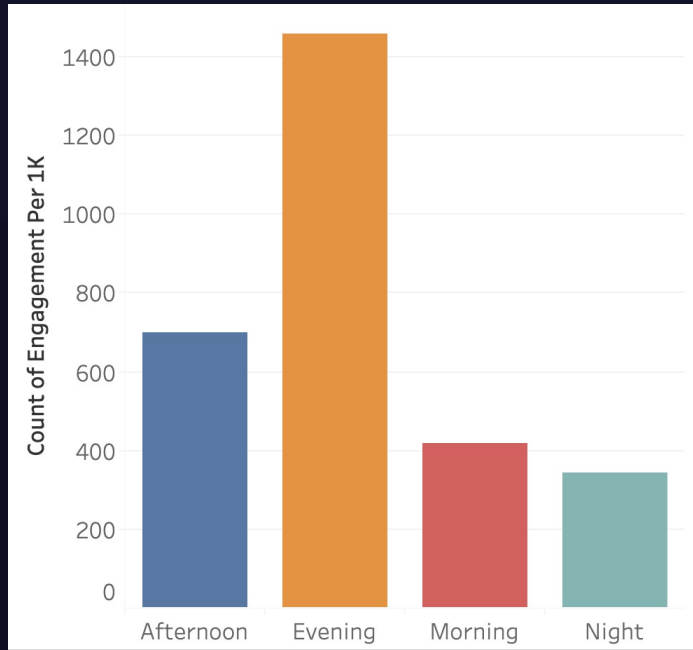
Limitations

1. In our dataset, everything varies at once. (e.g., "Night" post + "Trending Sound" + "Famous Creator").
1. Observational Data: Not controlled data

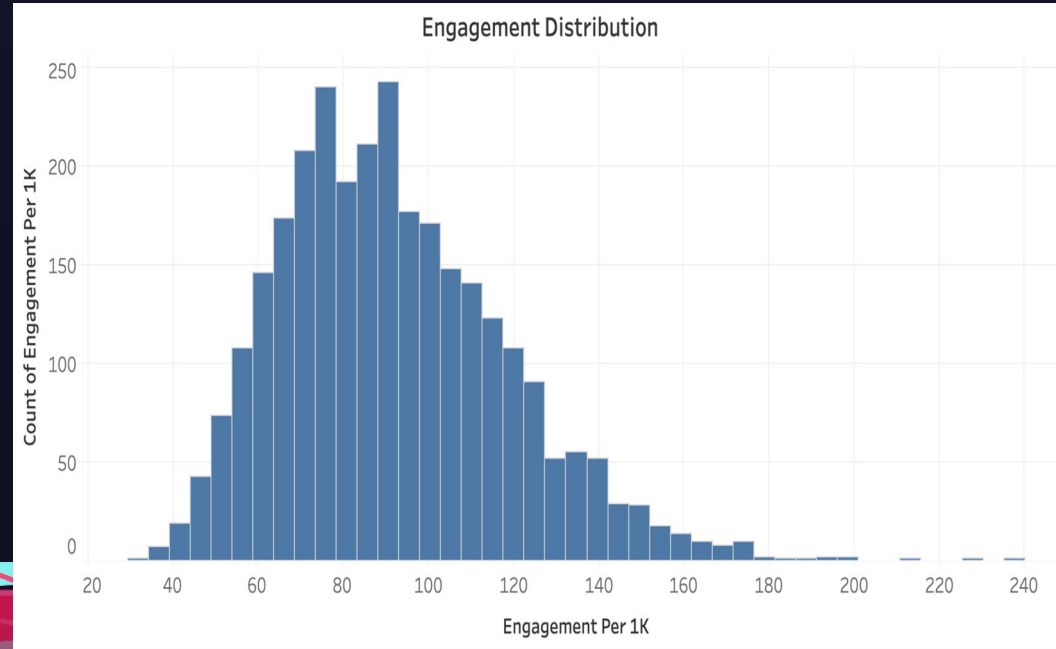


Descriptive Statistics

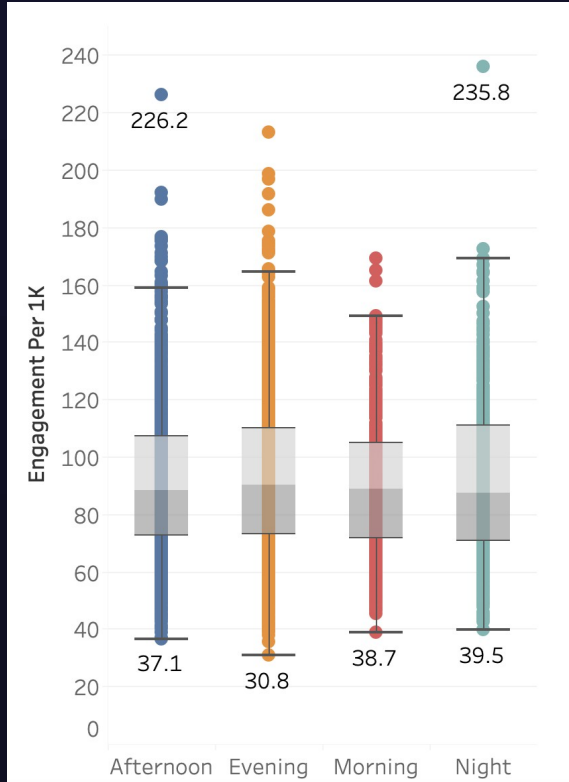
Videos Published Per Time Period:



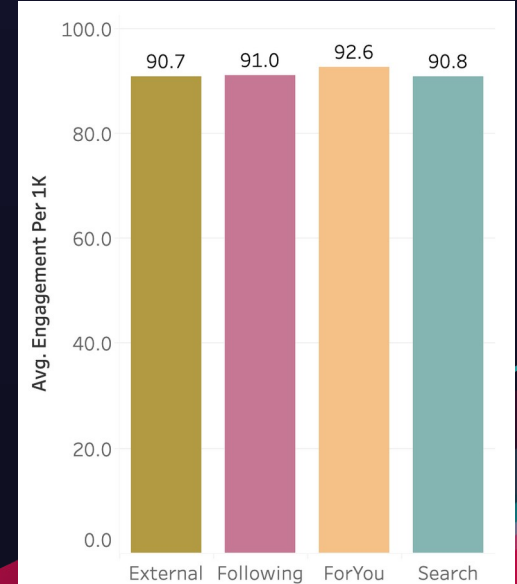
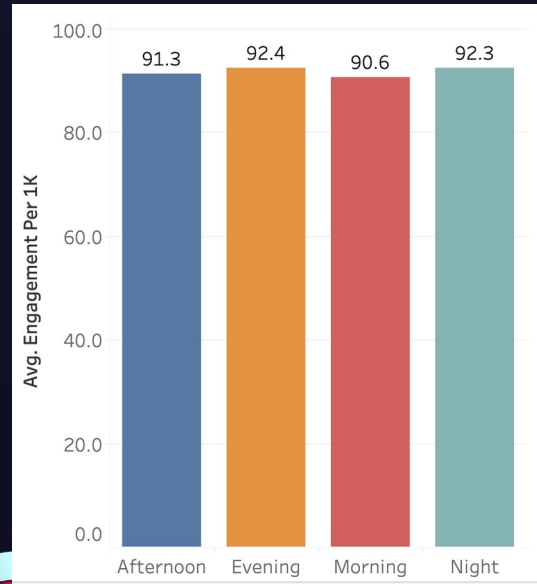
Engagement Distribution:



Descriptive Statistics



Stratified Dataset:



Regression Analysis

Summary of Fit

RSquare	0.000662
RSquare Adj	-0.00037
Root Mean Square Error	26.72583
Mean of Response	91.89088
Observations (or Sum Wgts)	2912

Analysis of Variance

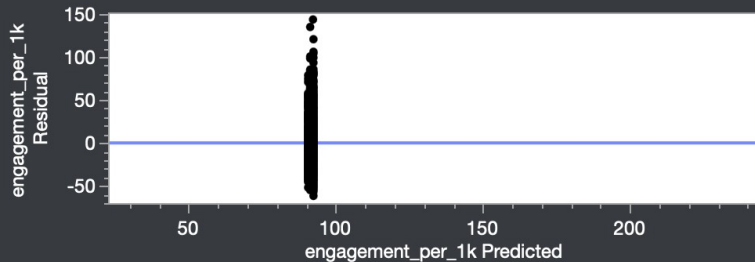
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	3	1375.8	458.615	0.6421
Error	2908	2077097.4	714.270	Prob > F
C. Total	2911	2078473.3		0.5879

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob> t	VIF
Intercept	92.434772	0.700648	131.93	<.0001*	
publish_period_Afternoon	-1.110566	1.230535	-0.90	0.3669	1.1250382
publish_period_Morning	-1.81128	1.484515	-1.22	0.2225	1.1023564
publish_period_Night	-0.155938	1.606056	-0.10	0.9227	1.0900008

Effect Tests

Residual by Predicted Plot



Interpretation

There is no relationship between the posting time and engagement.

Why Is R-Square Almost 0?

It is highly unpredictable to estimate human behaviors / whether a post would become 'viral'.

Key Takeaways: Deconstructing Engagement

No Relationship?

- Algorithm does not prioritize post timing
- Observational data limitations
- Linear Model may not be best fit
- Engagement distribution is dominated by outliers



For Creators:

- Time does not “hack” the algorithm

Summary



Hypothesis

Does posting in the evening influence TikTok engagement?



Regression

Posting in the evening has no impact on engagement.



Reasons

Human Behavior, Linear model is a poor fit



Further Steps

Take more variables into consideration, test out random forest model, consider quality of content

THANKS!

